

Khalil Chaghouri

Electrical & Computer Engineering Student | University of Toronto

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Summary

ECE undergraduate working across AI systems, EEG and brain-computer interfaces, and embedded hardware. Executive Officer at Mersivity (MannLab), co-author at IEEE GEM 2026, and builder of LLM-powered products for three companies in Kuwait. Open to Summer 2027 research and engineering roles.

Education

University of Toronto | *B.A.Sc., Electrical & Computer Engineering (PEY Co-op)*
American School of Kuwait | *Valedictorian*

Sep 2025 – Apr 2030
Jun 2025

Publications

Cocaine Buzzkill: An XR Game To Take Aim Against the Drug Epidemic. IEEE GEM 2026, Berlin (third of five authors).

Experience

Researcher, MannLab & Executive Officer, Mersivity | *University of Toronto*

Nov 2025 – Present

- Executive Officer since Sep 2026: lead student research projects and help define new research directions for the lab.
- Wrote the IEEE GEM 2026 paper (3rd of 5 authors) on an EEG-driven XR snowball game; helped debug its Unity app.
- Built a webcam piano-coaching system (MediaPipe, OpenCV) end to end and ran its pilot EEG experiments.
- Developing a quantum Adaptive Chirplet Transform in Qiskit: chirps encoded as phase gates, read out with one QFT.

Lead Product Engineer, AmanTec | *Cybersecurity & AI, Kuwait (Remote)*

Aug 2026 – Present

- One of the main engineers and product owners on an unreleased AI product; designed its database schema, LangGraph agent architecture and access-control model (FastAPI, PostgreSQL). Wrote ~440 tests; logged 45 defects pre-launch.

Agentic Product Engineer, IDRAX Training & Consultancy | *Kuwait (Remote)*

Aug 2026 – Present

- Built an AI trainer-matching platform where a panel of LLMs grades trainers against training opportunities and a judge model ranks the best fit (FastAPI, LangGraph, Next.js); Arabic-capable CV OCR, encryption at rest, 934 tests.

Agentic Development Consultant, Jenayen | *Salon group, Kuwait (Remote)*

Sep 2026 – Present

- Building a Django finance platform in which Claude reads English and Arabic invoices, even handwritten; 232 tests.

Electrical Team Member, Robotics for Space Exploration (RSX) | *Toronto, ON*

Sep 2025 – Present

- Hardware verification and electrical design for the team's Battery Management System (BMS).

Projects

Shape from Lighting | *Low-contrast computer vision (solo)*

Summer 2026

- Physics-informed U-Net that finds white-on-white objects from shading; 43% lower error than classical photometric stereo.

Butler[bot] | *Battle of the Schools hackathon, team of 4*

Sep 2026

- Built the LLM tool-calling agent, initial balancing/driving in MuJoCo and Whisper voice; team hit 28/30 chores.

Qubit Router | *Q-SITE 2026 quantum hackathon*

Sep 2026 – Present

- Built GNN + policy nets on a self-generated dataset to prune routing search; beats Qiskit SABRE (67.0 vs. 76.0).

KC-Bots: Self-Assembling Swarm Robots | *Team lead, Kuwait*

Summer 2025

- Custom KiCad ESP32 PCB (IMU, motor drivers, electromagnet), C++ firmware, ESP-NOW mesh, STL-to-voxel planner.

Todosive | *Personal productivity app (solo)*

2026

- Offline-first Svelte 5 + Fastify/SQLite app with an Electron client; ~300 MB RAM vs. 2 GB for my old Notion setup.

AI Scam Detector | *AP Research, Kuwait*

Sep 2024 – May 2025

- One-hotkey scam checker for older adults (Google Cloud Vision + ChatGPT); F1 of 89% across emails, websites and calls.

Technical Skills

Languages

Python, C, C++, TypeScript / JavaScript, Java, SQL

AI / ML

PyTorch, LangGraph, LLM APIs (Claude, Gemini, Mistral), YOLO, MediaPipe, OpenCV, Qiskit

Software

FastAPI, Django, Next.js / React, Svelte, PostgreSQL, Docker, Electron, GitHub Actions, Linux

Hardware / BCI

ESP32, KiCad, soldering, BMS verification, EEG (Muse, MNE-Python), MuJoCo, Fusion 360

Leadership & Activities

TEDxYouth Speaker, "Paths of Learning" (2024) • Founder, ASK Robotics Club • Ran the "STEM for Change" hackathon